

Multi-Stage Manufacturing for Preoperative Medical Models with Overhanging Components

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ABSTRACT

Medical models play an instrumental role in replicating a patient's unique anatomy, enhancing surgical outcomes, and fostering effective communication between doctors and patients. While 3D printing technology offers bespoke solutions for such models, producing designs with multiple overhanging tissues often requires advanced 3D printers – making it unattainable for conventional FDM fabrication methods. To bridge this gap, we present a cost-efficient, multi-stage, printing-molding hybrid manufacturing technique that incrementally solidifies complex medical models. Leveraging our adaptive optimization algorithm, we determine the optimal molding direction and the most streamlined manufacturing process with the fewest stages. As a testament to our method's efficacy, we successfully printed a liver model featuring overhanging tumors.

CCS CONCEPTS

• Computing methodologies → Shape modeling.

KEYWORDS

Medical preoperative model, 3D printing, Injection molding, Hybrid manufacturing

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1 INTRODUCTION

Medical models, particularly 3D printed ones, are vital for medical education, surgical preparation, demonstrations, and research. Compared to 2D images, they offer a realistic view of organs, enhancing comprehension, treatment planning, and surgical success rates. While 3D printing offers tailored solutions for medical modeling, attempts using advanced techniques like PolyJet for multi-material, multi-color preoperative models have been limited by their high costs [Chen et al. 2020].

To address the cost challenges in medical models' fabrication, Yang et al. segmented models into vascular and tissue structures.

Using mixed-color FDM 3D printing for vascular components and DLP for the shell, they achieved a high-quality transparent model by assembling with silicone, reducing costs [Yang et al. 2022]. Yet, some models have floating tissues like tumors, not connected to primary structures, and their accurate placement is vital for surgery planning.

To resolve this, we introduce a layered 3D printing method for models with overhanging tissues. The process involves analyzing the model, categorizing the tissues, and determining the best injection direction. Supports are added per layer to position overhanging parts on the shell. After printing and assembly, the model is finalized by infusing silicone gel and removing supports. Refer to Figure 1 for a visual overview.

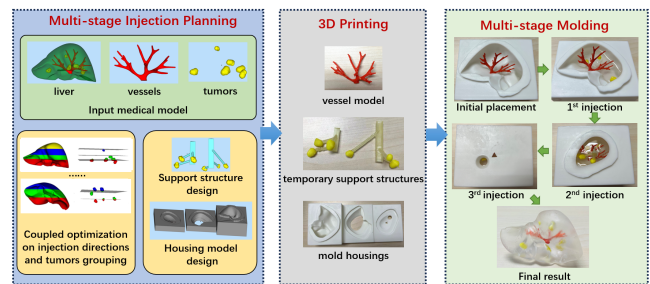


Figure 1: Framework of the proposed method.

2 THE PROPOSED METHOD

2.1 Problem formulation

Given a 3D medical model comprising three components - the vascular structure, overhanging tissues (such as tumors), and the remaining form - our objective is cost-effective fabrication. We merge 3D printing and molding methods. Specifically, the vascular structure and overhanging tissues are 3D printed independently, while the entire shape undergoes a phased injection molding process. The fabrication constraints are threefold:

1. All injection stages follow the same direction and the mold from each stage is removable.
2. The vascular structure is incorporated in the first molding phase.
3. Overhanging elements are supported by temporary structures, which are dissolved right after the specific molding step.

To optimize the phased injection molding with the fewest stages, we tackle this hybrid manufacturing challenge in two steps: multi-stage molding planning and accessories design. Molding planning focuses on pinpointing the ideal orientation for the sequence, while

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accessories design is concerned with creating the mold housings and the temporary support structures for the overhanging tumors.

2.2 Multi-stage injection planning

To accurately position the overhanging components during fabrication, we utilize a temporary, soluble support structure to anchor them within the mold. Post-molding, these structures are dissolved in water. It's imperative that no overhanging structure is fully submerged during each injection to ensure complete removal of these supports.

We've introduced an algorithm for multi-stage injection planning. It identifies an injection direction that groups overhanging components into the fewest clusters, taking into account the three aforementioned fabrication constraints. Grouping is guided by the rule that an overhanging structure can be integrated into a group if its height interval overlaps with the collective intersecting height of previously grouped structures. Given that the injection direction remains consistent across stages, our algorithm operates by examining each sampling direction. The process is outlined below.

Algorithm 1: Multi-stage Injection Planning

Input: Given medical model with overhanging tumors

Output: The injection direction $OptDir$ and tumor groups

```

1 foreach sampling manufacturing direction  $dir$  do
2   Sort all tumors by height position along  $dir$ ;
3   Cluster the tumors into injection groups;
4   if the vascular structure placement constraint is not
      satisfied then continue;
5   if the molding removal constraint is not satisfied then
      continue;
6    $OptDir \leftarrow$  number of groups fewer ?  $dir : OptDir$ ;
7 end
8 Return  $OptDir$  with the fewest number of groups.
```

2.3 Accessories design

2.3.1 Temporary support structures. For each injection, tumors of the same level are anchored using support structures to maintain their accurate positioning within the liver. These structures are consistently fixed atop the shell model across injection layers.

To prevent collisions with vascular structures, we employ an optimal voxel search method to auto-generate tree-like supports, comprising primary and secondary branches. The guiding principles of our voxel search are as follows: ensuring that the angles between secondary and primary branches are maintained at 45 degrees or below, avoiding intersections between all branches and vascular structures within the voxel space, and ensuring that all supports remain within the boundaries of the shell model.

2.3.2 Mold housings. The mold housing for each stage is derived from the horizontal segmentation of the shell model. To guarantee safe injection, the shell model's height surpasses the injection line. As a result, we set a height threshold to regulate the housing's stature. Moreover, for a flawless assembly of consecutive housings, we incorporate an embedding structure between them, reminiscent of the approach in [Yang et al. 2022].

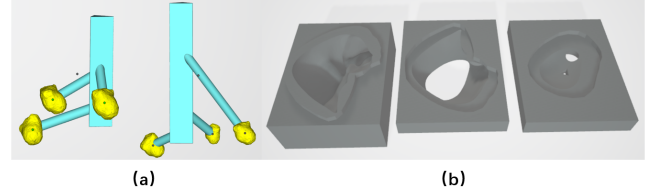


Figure 2: (a) Temporary support structures (blue) that connect the tumors (yellow) (b) The three mold housings.

3 EXPERIMENT AND RESULT

Following our algorithmic procedure, we have determined the optimal multi-stage molding plan and accessory designs for an input liver model with overhanging tumors. Figure 3 details our fabrication process. The shell mold is crafted using large-scale DLP technology [Wu et al. 2018], while the vascular and tumor models employ multi-material FDM technology [Wu et al. 2021]. Notably, the support portion of the overhanging tumor is printed with water-soluble material.

After printing these components, assembly ensues based on the optimal multi-stage molding plan. In this instance, achieving a comprehensive liver model with overhanging elements necessitated three injections.

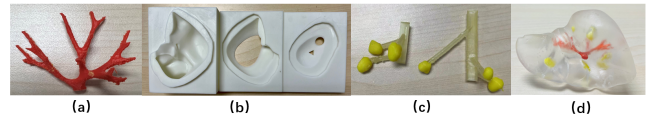


Figure 3: Illustration of the manufacturing process.

4 CONCLUSION

We introduce an efficient multi-stage manufacturing approach for intricate medical models. Our adaptive algorithm fine-tunes the molding direction, aiming to achieve the optimal manufacturing scheme with the fewest injections as its iterative target. While this method is demonstrated on liver models containing tumors, it holds broader applicability, extending to printing tasks for various human organs with irregularly placed abnormal tissues.

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